

Arham Shah

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SKILLS

Technical: SolidWorks (CAD), FEA, Structural Stress Analysis, MATLAB, C++, Thermal Loading, Thrust Frame Design, Injector Feed Lines, Flame Deflector Systems, Soldering

Operational: Technical Documentation, Research Presentation, Risk Assessment, Safety Protocols, Hazard Mitigation, Data Analysis, Project Management, Operations Management

EXPERIENCE

Longhorn Rocketry Association, Austin, TX

Structures Engineer

09/2025 - Present

- Engineered a vertical rocket engine test stand capable of managing 3 Kilonewtons of thrust by designing a high-load thrust frame, specialized injector feed lines, and a custom flame deflector system.
- Ensured structure and deflector safety factor requirements were met by conducting comprehensive Finite Element Analysis (FEA) and stress analysis under high thermal loading.

ACE Mentor Program of Dallas/Fort Worth, Frisco, TX

Engineering Consultant

10/2024 - 02/2025

- Executed a comprehensive design-build project by collaborating within a multidisciplinary engineering team and interfacing with architecture and construction professionals.
- Acquired practical project lifecycle expertise from schematic design to construction administration by analyzing immersive site visits and applying mentorship.

Young Jains of America/JAINA, Dallas, TX

College Chapters Team

10/2025 - Present

- Empowered university students by collaborating within a specialized team to execute targeted leadership initiatives.
- Maximized youth engagement across collegiate demographics by aligning programmatic goals with core cultural values and community outreach strategies.

Multimedia Team Specialist

01/2022 - Present

- Resolved 100+ end-user connectivity anomalies by engineering a virtual help desk and implementing troubleshooting protocols.
- Optimized broadcast reliability for events by configuring multimedia systems and actively monitoring real-time stream integrity.

Texas Athletics, Austin, TX

Student Associate - Game Operations

09/2025 - 12/2025

- Reduced event setup time by 15% for 100,000+ attendees by engineering equipment handling workflows and resolving constraints.
- Streamlined mission-critical operations for high-attendance events by applying systems-oriented problem-solving and real-time issue resolution frameworks.

Frisco Ignite, Frisco, TX

Media Co-Lead: Lead for Long Form Content

09/2022 - 05/2025

- Directed end-to-end execution of large-scale interactive sessions for attendees by standardizing operational procedures across multidisciplinary teams and speakers.
- Enhanced future programming efficacy by analyzing post-event feedback metrics and facilitating small group discussions to maximize learning outcomes.

RESEARCH PROJECTS

A Comparative Study of Airfoil Performance

Simulated Catapult System

09/2021 - 05/2025

- Improved aerodynamic design accuracy by 10% and optimized flight stability by conducting a comparative study of airfoil performance and analyzing lift-to-drag tradeoffs.
- shahphysics.weebly.com/aircraft-catapult.html

EDUCATION

The University of Texas at Austin

B.S. Aerospace Engineering

08/2025 - 12/2028

- **Relevant Coursework:** Engineering Computation, Statics, Applied Thermodynamics
- **Tools & Languages:** MATLAB, C++, Microsoft Office Suite, Data Analysis

CERTIFICATIONS

- Microsoft Office Specialist: Word Associate
- Microsoft Certified Educator
- Certiport: Intuit Personal Finance
- Forage: British Airways Data Science
- Google Cloud: Intro to Large Language Models
- Mathworks: MATLAB Onramp

Competitive GeoGuessr Player